Agents That Forget How to Chat: Self-Distilled Rehearsal Prevents Tool-Call Collapse Through SFT and RL Post-Training of a Shopping Agent

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Abstract

Post-training a general instruction-tuned model into a tool-using agent can silently destroy its ability to behave as a general assistant. We study this in a multi-turn, permission-gated shopping environment with hidden user constraints, fine-tuning Qwen2.5-7B-Instruct with LoRA, first by supervised fine-tuning (SFT) on expert demonstrations and then by RLOO reinforcement learning against a ground-truth reward. A specialist trained only on shopping data reaches 93.8 $\pm$2.7% task success on a held-out catalog after RL (3 seeds), but answers only 22.5 $\pm$9.0% of 40 general questions without collapsing into store actions (base model: 95.0%), and its IFEval prompt-level strict accuracy falls from 64.5% to 38.8 $\pm$0.8%. MMLU log-likelihood accuracy is unchanged ($\approx$74% in every arm): the model's knowledge survives while its behavior does not. Generative probes confirm this: under the deployment prompt the specialist answers 0.0% of TriviaQA and multilingual math (MGSM) questions correctly on every seed, yet with the system prompt removed the same post-RL checkpoints score 69.6 $\pm$1.2% and 51.1 $\pm$0.8%, matching the base model. Mixing 25% *self-distilled* rehearsal (open-ended Dolly prompts answered by the untrained base model) into SFT keeps both measures at or above the base model through SFT and RL (100.0% Retention-40, 65.5 $\pm$0.5% IFEval, 69.9 $\pm$0.2% TriviaQA under the deployment prompt). Task success is lower on average (83.9 $\pm$9.4% vs. 93.8 $\pm$2.7%) and more variable across seeds, although with three seeds this gap is not statistically significant (exact permutation test $p = 0.30$). We release code, configurations, per-run manifests and all raw evaluation outputs.

1 Introduction

A common recipe for building an LLM agent is to take an instruction-tuned model, fine-tune it on task demonstrations, and then improve it with reinforcement learning against a task reward. Deployed agents, however, rarely see only in-task traffic: a shopping assistant is also asked what the capital of Australia is. If post-training teaches the model to respond to *everything* with a store action, the agent becomes worse at the general assistance it was supposed to retain.

Catastrophic forgetting under sequential training is long known [7, 5] and has been measured in LLM continual fine-tuning [6]; replaying old data (rehearsal) is one of the oldest mitigations [10], and mixing pre-training data into RLHF updates is standard practice [8]. Closest to our method, self-distillation fine-tuning (SDFT) [13] reduces forgetting by having the seed model rewrite the task data in its own words before SFT. Our work differs in three respects. (i) *Setting*: we study a tool-using agent trained by SFT *followed by* RL, and ask whether the RL stage repairs, compounds or leaves unchanged what SFT did. (ii) *Failure mode*: the loss we measure is not degraded knowledge or safety but *tool-call collapse*, where the agent answers off-task questions with store actions; we show that a knowledge benchmark misses it entirely. (iii) *Mechanism and cost*: the task data is left untouched and the base model's answers to unrelated open-ended prompts are mixed in as a separate rehearsal stream, and we report what this costs in task success. This paper contributes a small, controlled, fully reproducible measurement:

- **Behavioral forgetting, not knowledge loss.** Shopping-only post-training collapses off-task chat into store actions and halves IFEval accuracy, while MMLU (scored by log-likelihood, independent of output format) is unchanged (§5). Removing the system prompt restores the specialist's TriviaQA and MGSM accuracy to base-model level: the collapse is a prompt-conditioned policy, not lost knowledge (§6).
- **RL neither causes nor repairs it.** For both arms, retention and IFEval after RL are close to their post-SFT values (within seed variation); the damage is set during SFT.
- **Self-distilled rehearsal prevents it, at a likely cost.** 25% rehearsal keeps Retention-40 and IFEval at base-model level on all three seeds. Post-RL task success is 9.9 points lower on average and more variable across seeds; with three seeds this gap is not significant ($p = 0.30$), so we report it as a likely cost rather than an established one.
- **A contamination-controlled evaluation.** We found that the environment's originally registered evaluation split shared 39 exact hidden needs and 52 answer sets with the training pools; all results here use a held-out catalog with need-level de-duplication (§4).

2 Environment

Task. Each episode is a multi-turn conversation (at most 12 turns) between the agent and a simulated shopper in a synthetic product catalog of 300 items. The shopper has a hidden budget and 3–4 hidden hard requirements that are revealed only in response to a relevant clarifying question. The agent acts by emitting one JSON action per turn (`ask_user`, `search`, `select_product`, `request_cart_permission`, `add_to_cart`) after a short natural-language sentence. User turns in our evaluation are Spanish–English code-switched.

Permission gate. `add_to_cart` for a product succeeds only if that exact product was previously granted by the shopper. The gate is enforced in the environment's state transition, so a violation cannot be rewarded.

Reward and success. The episode reward is $R = 0.4\,q + 0.4\,a + 0.2\,p - 1.0\,v + \sum_t g_t$, where q is the value quality of the purchased item against catalog ground truth, a indicates the shopper accepted the item, p that permission was requested, v a permission violation, and g_t a per-turn information-gain bonus for questions that eliminate inconsistent products. There is no learned reward model. An episode is a *success* if $q > 0$ and $v = 0$. We additionally report regret in US dollars against the cheapest item satisfying all hidden requirements.

3 Method

Model and adapters. All arms fine-tune Qwen2.5-7B-Instruct [9] with a rank-16 LoRA [4]; the base weights stay frozen and only the adapter is trained.

SFT. 540 expert demonstrations recorded programmatically in the environment on the training catalog (catalog seed 0), in mixed languages and including cases where the shopper denies permission or no product satisfies the needs, one epoch, learning rate 10^{-5}, batch size 2, maximum length 2048 tokens. The loss is computed on agent tokens only. The demonstration seed is fixed across runs; the training seed controls initialization and batch order.

Self-distilled rehearsal. We select 900 Dolly-15k [2] prompts from the open-QA, general-QA, brainstorming and creative-writing categories that have no context passage, after filtering out shopping-related words, and answer each with the *untrained* base model under a plain helpful-assistant system prompt. Answers that hit the generation limit without an end-of-sequence token are dropped (57 of 900), leaving 843 exemplars. The rehearsal arm adds the first 180 exemplars to the 540 shopping demonstrations, so rehearsal is 25% of the SFT mixture. Because targets come from the same base model, rehearsal asks the adapter to *preserve* existing behavior rather than to learn new content.

RL. Starting from each arm's SFT adapter, we run 50 steps of RLOO [1] with $k = 8$ rollouts per prompt from a pool of 64 training scenarios, learning rate 10^{-6}, and a per-token k3 KL penalty [11] ($\beta = 0.04$) to the SFT adapter. The advantage of rollout i is $R_i - \frac{1}{k-1}\sum_{j \neq i} R_j$. The policy and the reference are two LoRA adapters on one shared frozen base, which fits a 7B policy-plus-reference on one 32 GB V100. RL uses shopping data only for both arms; rehearsal enters only through SFT.

Arms. *Specialist*: SFT on shopping only, then RL. *Rehearsal25*: SFT on the 75/25 mixture, then the identical RL stage. Each arm is run with training seeds 1, 2, 3 and evaluated after SFT and after RL.

4 Evaluation

Contamination audit and held-out split. The environment's previously registered evaluation split (scenario seed 5001, same catalog as training) shares 39 exact hidden needs and 52 exact answer sets with the SFT and RL pools; we verified this with a unit test. All shopping results here instead use 128 scenarios on a catalog generated with an unseen seed (1), dropping any scenario whose hidden need also appears in training. Every run is evaluated on the same 128 scenarios.

General-ability metrics. We measure three things, chosen to separate *knowing* from *behaving*:

- **Retention-40**: 40 fixed general questions (27 English, 13 Spanish; disjoint in template from the rehearsal prompts), asked under the deployment system prompt with greedy decoding. A response counts as retained if it is natural-language prose of at least 8 characters and does not parse as a store action. This measures collapse into the agent format, not answer correctness.
- **IFEval** [14] prompt-level strict accuracy on the first 200 prompts, with the chat template and the deployment system prompt (generative; behavior).
- **MMLU** [3], 0-shot, log-likelihood scoring, 10 items per subject (570 items; knowledge, independent of output format).

Both IFEval and MMLU use lm-evaluation-harness 0.4.5 [12].

Behavior and correctness probes. To separate *whether* the model answers from *whether it is right*, and to test what the collapse depends on, we run a second greedy evaluation on every SFT and RL checkpoint of both arms and on the base model (`scripts/paper/eval_behavior.py`; every generated answer is saved).

- **Off-task tool rate**: the fraction of 240 non-shopping prompts (the 40 Retention-40 questions plus 200 held-out Dolly prompts, asserted disjoint from the rehearsal set) answered with a parseable store action. It is measured under four system prompts: the deployment prompt; the deployment prompt plus an explicit instruction to answer non-shopping questions directly in prose (*deploy+direct*); a generic helpful-assistant prompt; and no system prompt.
- **Answer correctness**: 300 TriviaQA questions (no-context setting; correct if any normalized gold alias appears as a whole-word span of the reply) and 300 MGSM grade-school math problems, 50 each in Spanish, French, German, Russian, Chinese and Japanese (correct if the last number in the reply equals the answer), under the deployment prompt and with no system prompt, at most 256 new tokens. A reply that is a store action is scored as wrong.
- **Adapter-scale sweep**: the LoRA update is multiplied by $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ ($\alpha = 0$ is the base model) and the tool rate is measured on the 40 Retention-40 questions under the deployment prompt.

We report the mean and sample standard deviation over three seeds and, per run, Wilson 95% intervals on task success. Arm comparisons use a Welch t-test and an exact permutation test on the three per-seed values (`scripts/paper/stats.py`).

Table 1: Main results on the held-out catalog (128 scenarios). Mean ± sample std over 3 training seeds; the base model is a single deterministic run. Permission violations were 0.0% for every trained run. All values are percentages except regret.

		Shopping		General ability		
Arm	Stage	Success ↑	Regret $ ↓	Retention-40 ↑	IFEval ↑	MMLU
Base 7B	–	0.0	–	95.0	64.5	74.0
Specialist	SFT	64.8 ±7.5	0.61 ±1.06	19.2 ±7.2	39.7 ±0.8	73.4 ±0.1
	SFT→RL	**93.8 ±2.7**	0.00 ±0.00	22.5 ±9.0	38.8 ±0.8	73.5 ±0.2
Rehearsal25	SFT	51.6 ±30.4	0.63 ±1.09	**100.0 ±0.0**	**66.0 ±0.5**	74.0 ±0.4
	SFT→RL	83.9 ±9.4	0.03 ±0.06	**100.0 ±0.0**	65.5 ±0.5	74.0 ±0.4

5 Results

The base model cannot do the task. Under the deployment system prompt the untrained model never produces a valid store action on this split (0% success, 100% invalid actions), so all task success in Table 1 comes from post-training.

Shopping-only post-training causes tool-call collapse. The specialist answers only 19.2% of general questions as prose after SFT and 22.5% after RL, versus 95.0% for the base model. The failures are qualitatively uniform: the model greets the user and emits a store action, for example answering "What's the capital of Australia?" with a greeting followed by `{"action": "ask_user", ...}`. Retention is lower for Spanish questions (15.4% vs. 37.0% English for specialist seed 1 after RL; single seed). IFEval drops by roughly 25 points.

Knowledge is intact. MMLU accuracy is 73.4–74.0% in every arm and stage, against 74.0% for the base model. Because MMLU is scored by log-likelihood over answer options and is insensitive to output format, this indicates that what is lost is the model's *policy for when to act as an assistant*, not the underlying knowledge. Practically, this means knowledge benchmarks alone would have certified the specialist as unharmed.

RL does not change the measured general behavior. For each arm, post-RL retention changes by at most 3.3 points and IFEval by at most 0.9 points relative to post-SFT, both comparable to the seed standard deviation. In our setting (50 steps, KL-regularized to the SFT adapter, KL around 5×10^{-4}–9×10^{-4} per token in the logs we inspected), the RL stage improves the task without moving Retention-40 or IFEval in either direction.

Rehearsal preserves Retention-40 and IFEval. With 25% rehearsal, retention is 100.0% on every seed and stage, and IFEval is 66.0 ±0.5% after SFT and 65.5 ±0.5% after RL, slightly above the base model's 64.5%. Rehearsal targets are the base model's own answers, so this is preservation rather than improvement; we do not interpret the 1-point difference.

The cost: lower and more variable task success. After RL, rehearsal25 reaches 83.9% success against 93.8% for the specialist. Table 2 shows the variance comes from individual seeds. After SFT, rehearsal seed 3 succeeded on only 16.4% of episodes, against 68.8–69.5% for seeds 1 and 2. That run never violated the permission gate, and every purchase it made was optimal (zero regret); the remaining episodes ended without a qualifying purchase (mean 11.8 turns vs. 11.1 for seed 1). RL raised this seed to 74.2%, but it remains the weakest run. Across seeds the post-RL success gap is not statistically significant (Welch t-test $p = 0.20$; exact permutation test $p = 0.30$), whereas the gaps in Retention-40 ($p = 0.004$) and IFEval ($p < 10^{-5}$) are (Welch). With three seeds per arm the exact permutation test cannot go below $p = 0.10$; it reaches that floor for both general metrics at both stages.

Table 2: Per-seed shopping success (%) with Wilson 95% intervals ($n = 128$ each).

Arm	Stage	Seed 1	Seed 2	Seed 3
Specialist	SFT	68.8 [60.3, 76.1]	69.5 [61.1, 76.8]	56.3 [47.6, 64.5]
Specialist	RL	92.2 [86.2, 95.7]	96.9 [92.2, 98.8]	92.2 [86.2, 95.7]
Rehearsal25	SFT	69.5 [61.1, 76.8]	68.8 [60.3, 76.1]	16.4 [11.0, 23.8]
Rehearsal25	RL	93.0 [87.2, 96.3]	84.4 [77.1, 89.7]	74.2 [66.0, 81.0]

Table 3: Behavior and correctness probes (%). Off-task tool rate is over 240 non-shopping prompts; TriviaQA $n = 300$; MGSM $n = 300$ (6 languages). Mean ± sample std over 3 training seeds; the base model is a single deterministic run.

		Off-task tool rate ↓			TriviaQA ↑		MGSM ↑	
Arm	Stage	deploy	deploy+direct	none	deploy	none	deploy	none
Base 7B	–	9.2	5.4	0.0	35.7	68.3	7.0	48.3
Specialist	SFT	86.7 ±7.8	61.4 ±8.8	0.0	0.0	69.6 ±1.4	0.0	52.0 ±0.9
	SFT→RL	86.5 ±10.3	60.7 ±6.9	0.0	0.0	69.6 ±1.2	0.0	51.1 ±0.8
Rehearsal25	SFT	0.1 ±0.2	0.1 ±0.2	0.0	70.0 ±0.0	70.8 ±0.5	53.8 ±0.8	52.4 ±1.4
	SFT→RL	0.1 ±0.2	0.1 ±0.2	0.0	69.9 ±0.2	70.7 ±0.3	53.9 ±1.3	53.4 ±0.8

6 What the Collapse Depends On

The specialist cannot answer under the deployment prompt. Under the deployment prompt the specialist answers 0.0% of TriviaQA and MGSM questions correctly on all six checkpoints (Table 3). About 87% of these replies are store actions; the remainder score zero as well, and those we read redirect the user to shopping ("I'm here to help with PennyMart stuff! ...") rather than attempt an answer. On the 240 off-task prompts, 86.7% (SFT) and 86.5% (RL) of replies are store actions, against 0.1% for rehearsal25 (Welch $p = 0.003$ after SFT, $p = 0.005$ after RL).

The knowledge is still there. With the system prompt removed, the same specialist checkpoints make no store actions and score 69.6 ±1.2% on TriviaQA and 51.1 ±0.8% on MGSM after RL, against 68.3% and 48.3% for the base model. Together with the unchanged MMLU scores, this shows that post-training did not remove the knowledge: it learned to withhold it whenever the deployment prompt is present. Under a generic helpful-assistant prompt the tool rate is also 0.0% for every checkpoint. An explicit instruction appended to the deployment prompt to answer non-shopping questions directly lowers the specialist's off-task tool rate only to about 61%, so a prompt-level fix does not remove the behavior.

The collapse grows smoothly with the adapter. Scaling the specialist's LoRA update by $\alpha = 0, 0.25, 0.5, 0.75, 1$ gives tool rates on the Retention-40 questions of 5.0, 14.2, 38.3, 63.3 and 82.5% after SFT and 5.0, 14.2, 38.3, 62.5 and 78.3% after RL (means over three seeds; the rate increases with α on every seed). For rehearsal25 the rate is 5.0% at $\alpha = 0$ (the base model) and 0.0% at every $\alpha > 0$ on every seed.

Rehearsal also preserves correctness. After RL, rehearsal25 checkpoints score 69.9 ±0.2% TriviaQA and 53.9 ±1.3% MGSM under the deployment prompt, matching their no-prompt scores. We also observe that the *untrained* base model itself scores lower under the deployment prompt (35.7% TriviaQA, 7.0% MGSM) than without it (68.3% and 48.3%). Reading its answers, we see it often decline off-topic questions as outside the shopping role or ask a clarifying question instead of solving the problem. The rehearsal checkpoints do not show this drop. A plausible explanation is that the rehearsal targets were generated under a plain assistant prompt, but this rests on a single deterministic base-model run and we have not tested it.

7 Status of the Results

This manuscript is a snapshot of an ongoing study and reports only completed experiments.

- **Confirmed (3 training seeds, every run evaluated):** all results in Tables 1–3; the Retention-40 and IFEval differences between arms; the off-task tool-rate and correctness differences under the deployment prompt; that MMLU and no-prompt correctness are unchanged by post-training; that RL leaves the measured general behavior unchanged.
- **Preliminary:** the task-success cost of rehearsal (9.9 points, not significant with three seeds); the base model's lower correctness under the deployment prompt and its explanation (one run); the English–Spanish retention gap (one seed).
- **In progress and not reported here:** other rehearsal doses (10% and 50%); rehearsal with human-written instead of self-distilled answers; KL-to-base regularization during SFT as an alternative mitigation; training seeds 4 and 5; and replication on a second model family (Mistral-7B-Instruct-v0.3). No results from these runs appear in this version.

8 Limitations

- **One model, one task, one rehearsal dose.** All results are for Qwen2.5-7B-Instruct with LoRA on a synthetic environment with a simulated shopper. We report a single rehearsal fraction (25%); other doses and a second model family are in progress and not reported here (§7).
- **Three seeds.** Seed variance in the rehearsal arm is large (std 30.4 points after SFT), so mean differences in task success should be read with caution.
- **Shared evaluation scenarios.** All runs use the same 128 held-out scenarios; seeds vary training, not the test set. The Wilson intervals reflect scenario sampling for one run, not training variance.
- **Subsampled general benchmarks.** MMLU uses 570 items and IFEval 200 prompts; small differences ($\leq$1 point) are below resolution.
- **Retention-40 is a format metric.** It checks that the model answers in prose rather than with a store action; it does not grade answer correctness. TriviaQA and MGSM (§6) cover correctness for short factual and arithmetic answers only; MGSM replies are capped at 256 new tokens, which can truncate long solutions in every arm alike.
- **No comparison to other mitigations yet.** This version does not compare against KL-to-base during SFT or rehearsal with human-written answers (both in progress), or against weight-space regularization.
- **Short RL.** 50 RL steps with a small KL penalty; longer or more aggressive RL may interact with forgetting differently.

9 Conclusion

Turning a chat model into a tool-using agent through SFT and RL can make it answer general questions with tool calls while leaving its knowledge intact, a failure that knowledge benchmarks do not detect. In our setting, the damage is done during SFT, RL leaves it unchanged, and a 25% mixture of the base model's own answers to open-ended prompts prevents it on every seed as measured by Retention-40, IFEval and answer correctness under the deployment prompt. Removing the system prompt restores the specialist's correctness, so the knowledge survives and only the policy for using it changes. Task success was 9.9 points lower on average, a gap our three seeds cannot distinguish from seed variance. Agent post-training pipelines should report behavioral, not only knowledge, retention.

Reproducibility. Code, the exact rehearsal prompt list, SLURM scripts, per-run manifests (command line, git commit, host, library versions) and all raw evaluation outputs are in the repository (`paper/results/raw/`); every number in this paper is generated from them by `scripts/paper/make_table.py` and `scripts/paper/behavior_table.py`. This snapshot is git tag `snapshot-2026-10-14`. Repository: https://github.com/parammadan/pennypilot (branch `paper/rehearsal`).

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A Training on Volta GPUs

All runs use NVIDIA V100-SXM2-32GB GPUs, which lack native bf16 support and FlashAttention. We train in fp16 autocast with dynamic loss scaling (GradScaler) and fp32 trainable LoRA weights. Sharing one frozen base between the policy and reference adapters avoids loading a second 7B model for the KL term. RL steps took roughly 160–230 s each in the logs we inspected; per SLURM accounting, one rehearsal-arm SFT stage including its evaluation took about 1.3 h and one RL stage including its evaluation about 3.7 h. In total, the GPU jobs for this paper used 33.1 V100 GPU-hours by SLURM accounting, including smoke tests and cancelled jobs.

B Additional baselines

On the same held-out split, Qwen2.5-1.5B-Instruct reaches 2.3% success (5.5% with shuffled search results), and Qwen2.5-7B-Instruct with an earlier, more prescriptive system prompt reaches 1.6%, confirming that the task is not solvable zero-shot. General-ability metrics were measured only for the base 7B model under the deployment prompt (Table 1).